Symmetrical Components & Power Transmission Line Faults

Fortescue transformation for positive/negative/zero sequences, single line-to-ground (SLG) fault calculation, and synchronous swing equation.

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1. Fortescue Symmetrical Components

Unbalanced three-phase phasors are resolved into Positive (a-b-c), Negative (a-c-b), and Zero sequence components.

2. Transmission Fault Analysis

Single line-to-ground fault current If = 3 * Ea / (Z1 + Z2 + Z0 + 3*Zf), modeled by series connection of the sequence networks.